Project Design Phase
Solution Architecture

Date	8-04-2026
Team ID	SWTID-2026-2088
Project Name	Automated Data Import and Relationship mapping using input sets & Dot walking.
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Design a module to import structured input sets (CSV, Excel, API feeds) into a staging area, ensuring validation, normalization, and error handling before processing.
- Implement logic to transform incoming data into target schema using configurable rules, including automated field mapping and relationship identification.
- Build a dynamic mechanism to traverse related records (dot walking) across tables to establish and validate relationships (e.g., reference fields, foreign keys).
- Store processed data into target tables with audit logs, while providing monitoring dashboards and error tracking for traceability and maintenance.

Example - Solution Architecture Diagram:

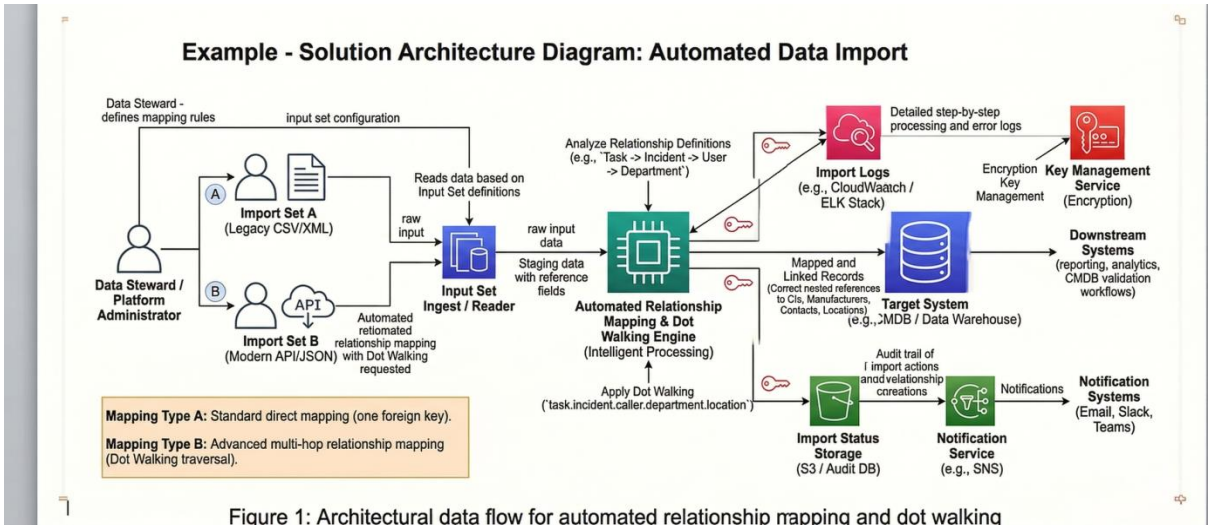

Figure 1: Architectural data flow for automated relationship mapping and dot walking

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/